Law as Extended Phenotype: A Game-Theoretic Microfoundation for Memetic Legal Evolution

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Abstract

This paper develops formal microfoundations for understanding legal systems as extended phenotypes (Dawkins, 1982). By "microfoundations" I mean the specification of agent-level mechanisms that produce macro-level patterns: rather than merely asserting that legal norms "evolve," this framework identifies precisely what is being selected, through what mechanism, and under what conditions selection produces stable outcomes.

Legal doctrines function as memetic strategies competing in repeated social games. Drawing on Maynard Smith and Price (1973) and Weibull (1995), I argue that legal norms can be analyzed as Evolutionarily Stable Strategies (ESS), subject to stability conditions determining persistence or collapse. The framework is operationalized through the Cultural-Legal Index 2.0 (CLI 2.0), measuring transmission capacity across three dimensions. Multi-AI inter-rater reliability testing yields ICC(2,1) = 0.9335 (n = 30 U.S. constitutional provisions).

CLI 2.0 discriminates between ESS provisions (mean = 0.812, n = 25) and collapsed provisions (mean = 0.521, n = 5), with $t = 6.73$, $p < 0.001$. Prohibition's CLI 2.0 of 0.303 (4.2 SD below mean) accurately predicted its vulnerability to invasion despite formal constitutional entrenchment.

The capacity gap hypothesis ($\Delta = TCI - CLI$) provides theoretical foundations for the environmental compatibility construct tested empirically in the IusMorfos model (Lerer, 2025), which empirically tested that environmental compatibility weighs 1.7× more than ideological commitment in determining institutional fitness (retrospective validation error: 12.4%, within 90% CI). The present paper explains why environmental compatibility dominates: target systems lacking transmission infrastructure cannot sustain high-complexity doctrines regardless of elite preferences.

The paper reconceptualizes institutional design as phenotype engineering: constructing legal memes whose extended phenotypic effects render desired behaviors evolutionarily stable.

Keywords: Extended Phenotype Theory, Evolutionary Game Theory, Evolutionarily Stable Strategies, Memetic Evolution, Legal Transplants, Constitutional Entrenchment, Cultural Transmission, Institutional Design, Comparative Constitutional Law

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I. Introduction: The Problem of Legal Microfoundations

1.1 The Evolutionary Analogy and Its Discontents

The application of evolutionary concepts to legal phenomena has a distinguished intellectual pedigree extending back to at least Maine's *Ancient Law* (1861) and reaching sophisticated expression in Holmes's aphorism that the life of the law has not been logic but experience. Contemporary scholars routinely invoke evolutionary language: legal doctrines "adapt" to changing circumstances, "selection pressures" favor certain institutional arrangements over others, precedents "survive" or "die," and legal systems exhibit "fitness" relative to their social environments. The intuition underlying such language is powerful: legal institutions, like biological organisms, appear to undergo processes of variation, selection, and inheritance that produce complex adaptive structures over time.

Yet this evolutionary metaphor, for all its intuitive appeal, has remained precisely that: a metaphor. Unlike evolutionary biology, which rests on precise microfoundations specifying the mechanisms of inheritance, variation, and selection, evolutionary legal theory has typically operated at a level of abstraction that renders its claims unfalsifiable. When we say that a legal doctrine has been "selected" for its "fitness," what exactly is being selected? What constitutes the relevant fitness function? Through what mechanism does selection operate? And crucially, what predictions does the evolutionary framework generate that would distinguish it from alternative explanations of legal change?

The absence of rigorous microfoundations has exposed evolutionary legal theory to devastating critique. The term "microfoundations" originates in economic methodology, where it denotes the derivation of aggregate phenomena from individual-level mechanisms. A macroeconomic model has microfoundations when it specifies how the decisions of individual agents, responding to incentives and constraints, produce the observed macro-level patterns. Without such specification, a model describes but does not explain; it identifies correlations but not causes.

Applied to legal evolution, the microfoundational challenge is this: if we claim that legal norms are "selected," we must specify what exactly is being selected (the unit of selection), by what mechanism (the selection process), and with what consequences

(the equilibrium conditions). Merely observing that some legal doctrines persist while others disappear does not constitute an evolutionary theory any more than observing that some firms survive while others fail constitutes an economic theory. The observation demands explanation, and explanation requires mechanism.

This paper provides the missing mechanism. Drawing on evolutionary game theory, I show that legal norms can be modeled as strategies in repeated social games, that legal systems reconfigure payoff structures to alter strategic equilibria, and that the persistence or collapse of legal arrangements depends on whether they satisfy the formal conditions for Evolutionarily Stable Strategies developed by Maynard Smith (1982). These conditions, not formal amendment rules or normative desirability, determine institutional fitness.

1.2 The Extended Phenotype Thesis

Dawkins's extended phenotype concept emerged from a fundamental insight about the unit of selection in evolutionary biology. Classical Darwinian theory focused on the organism as the unit of selection, with genes serving merely as the mechanism of inheritance. Dawkins inverted this perspective: genes are the true replicators, and organisms are survival machines, vehicles constructed by genes to ensure their own propagation. But if we take the gene's-eye view seriously, there is no principled reason to restrict phenotypic effects to the boundaries of the organism's body. A gene may be selected for any systematic effect it produces in the world, whether that effect is expressed within the organism or outside it.

The paradigmatic examples are compelling. The beaver dam is not part of the beaver's body, yet genes for dam-building are as much subject to natural selection as genes for tooth enamel or tail shape. The dam is an extended phenotype: a structure in the external world whose properties affect the replicative success of the genes that code for dam-building behavior. Similarly, spider webs, bird nests, and caddisfly cases are all extended phenotypes, material artifacts through which genes reach beyond the body to manipulate the environment in fitness-enhancing ways.

Dawkins extended the concept further to encompass manipulation of other organisms' behavior. Parasites that alter host behavior to facilitate their own transmission are expressing extended phenotypes in the behavior of their hosts. The lancet fluke *Dicrocoelium dendriticum* induces ants to climb to the tops of grass blades where they are more likely to be eaten by sheep, the fluke's definitive host. The ant's behavior is part of the fluke's extended phenotype.

The transposition to law is natural once we recognize cultural replicators, memes, as analogous to genes. A legal doctrine is a replicable unit of information that propagates through populations via judicial decisions, statutory enactments, scholarly commentary, legal education, and professional socialization. Just as a gene codes for phenotypic effects that influence its own replication, a legal meme codes for behavioral

strategies and institutional arrangements that influence its own transmission. The courthouse, the law school curriculum, the treatise, and the compliance routine are all extended phenotypes of legal memes: structures in the social world that systematically affect the replicative success of the doctrines that generate them.

1.3 The Game-Theoretic Turn

The extended phenotype framework provides the conceptual vocabulary for understanding law as a replicating system, but it does not by itself specify the selection mechanism. In biological evolution, the selection mechanism is differential reproductive success: organisms with fitness-enhancing traits leave more offspring, and those offspring inherit the traits. What is the analogous mechanism for legal memes?

The answer lies in evolutionary game theory. A legal norm can be modeled as a *strategy* in a repeated social game, a rule of behavior that agents execute in their interactions with one another. Judges apply precedent. Citizens comply with property rules. Bureaucrats enforce regulations. Lawyers advise clients based on doctrinal expectations. These are all strategic behaviors coded by legal memes and executed in the context of ongoing social interactions.

The key insight is that legal systems do not merely regulate behavior; they *reconfigure the payoff structure* of social games. A liability rule changes the expected utility of tortious conduct. An anticorruption regime alters the risk-reward calculus of bribery. A judicial review doctrine transforms the equilibrium of legislative-executive bargaining. This is the extended phenotypic effect of legal memes: they modify the strategic landscape in which agents operate, creating new equilibria and destabilizing others.

Once we recognize legal norms as strategies in repeated games, the machinery of evolutionary game theory becomes applicable. A legal norm is *evolutionarily stable* if it cannot be invaded by mutant strategies; that is, if a population of agents following the norm cannot be successfully infiltrated by agents following deviant strategies. The formal conditions for evolutionary stability, developed by Maynard Smith and Price (1973) and refined by subsequent theorists, provide precise criteria for determining when a legal arrangement will persist and when it will be displaced.

1.4 Overview and Contributions

This paper develops the game-theoretic microfoundation for law as extended phenotype through several interconnected contributions:

First, I provide formal definitions of legal memes, legal strategies, and extended legal phenotypes that render the evolutionary analogy precise and operationally meaningful. These definitions establish the conceptual infrastructure for applying evolutionary game theory to legal phenomena.

Second, I adapt the canonical ESS framework to legal contexts, specifying the conditions under which legal norms exhibit invasion resistance, maintain polymorphic equilibria with controlled non-compliance, or become locked into coordination equilibria that may be suboptimal but path-dependent.

Third, I develop and test a novel measurement instrument, the Cultural-Legal Index 2.0, that quantifies the transmission capacity of legal memes. Drawing on Colagè and d'Errico's (2025) framework for analyzing evolutionary pathways in cultural phenomena, CLI 2.0 operationalizes transmission capacity across three dimensions: spatial, temporal, and social transmission. Multi-AI inter-rater reliability testing establishes excellent measurement properties (ICC = 0.9335).

Fourth, Empirical analysis indicates that CLI 2.0 discriminates between legal provisions exhibiting ESS properties and those that have collapsed or been repealed, with the paradigmatic case of Prohibition illustrating transmission infrastructure failure.

Fifth, I develop the capacity gap hypothesis for legal transplants, showing that transplant success depends on the relationship between source doctrine complexity (CLI 2.0) and target system transmission infrastructure (TCI), and I illustrate this with comparative case studies.

Sixth, I reconceptualize institutional design as phenotype engineering, deriving operational principles for creating legal arrangements whose extended phenotypic effects render desired behaviors evolutionarily stable.

The paper proceeds as follows. Section 2 develops the formal framework, defining legal memes, strategies, and extended phenotypes, and adapting ESS concepts to legal contexts. Section 3 presents the Cultural-Legal Index 2.0 and its validation. Section 4 reports empirical findings on ESS discrimination. Section 5 analyzes coordination games, polymorphic equilibria, and lock-in dynamics. Section 6 develops the legal transplant framework and institutional design principles. Section 7 discusses implications and limitations. Section 8 concludes.

II. Formal Framework: Legal Norms as Strategies in Evolutionary Games

2.1 Foundational Definitions

The formal framework begins with precise definitions of the core concepts. These definitions are not merely stipulative; they are designed to preserve the essential features of both legal theory and evolutionary biology while enabling rigorous analysis.

Definition 1 (Legal Meme). A *legal meme m* is a replicable unit of legal information capable of being transmitted across agents via legal education, judicial opinions, statutory enactments, enforcement routines, scholarly commentary, and professional socialization. Examples include: doctrines (substantive due process, the rule against

perpetuities), precedents (*Marbury v. Madison*, *Brown v. Board of Education*), statutory provisions (Section 1983, the Commerce Clause), and institutional practices (stare decisis, judicial review).

The definition emphasizes *replicability*: a legal meme must be capable of accurate transmission from one host to another. This distinguishes legal memes from idiosyncratic judicial preferences or unique historical events that do not propagate. The channels of transmission, legal education, judicial citation, legislative borrowing, and scholarly commentary, constitute the reproductive machinery through which legal memes replicate.

Definition 2 (Legal Strategy). A *legal strategy s* is a rule of conduct that agents execute in social interactions, either consciously (contract negotiation, litigation decisions) or through habituation (complying with traffic rules, respecting property boundaries). Legal memes encode strategies: the relationship $m \rightarrow s$ maps legal information to behavioral dispositions.

The meme-strategy mapping is crucial. A legal meme does not directly cause behavior; rather, it codes for a strategic disposition that agents express in appropriate contexts. The same meme may generate different strategies in different agents depending on their roles: the doctrine of consideration generates one strategy for contract drafters (include nominal consideration), another for litigators (challenge contracts lacking consideration), and yet another for judges (enforce only contracts supported by consideration). But all these strategies derive from the same underlying meme.

Definition 3 (Extended Legal Phenotype). The *extended phenotype* of a legal meme *m* is the set of all systematic effects it produces on the institutional environment beyond the immediate behavioral dispositions of individual agents. These include:

- *Material structures*: courthouses, prisons, regulatory agencies, law enforcement infrastructure, compliance departments
- *Informational artifacts*: precedent databases, treatises, casebooks, restatements, model codes, legal memoranda
- *Behavioral routines*: compliance protocols, enforcement procedures, adjudicatory processes, professional norms
- *Doctrinal scaffolding*: interpretive canons, balancing tests, burden-shifting frameworks, standards of review

The extended phenotype concept captures a fundamental insight: legal memes succeed or fail based not only on the behaviors they induce but on the entire infrastructure they construct in the institutional environment. A doctrine that generates supportive precedents, spawns specialized enforcement agencies, becomes embedded in law school curricula, and develops elaborated doctrinal frameworks has constructed an extended phenotype that facilitates its own replication. A doctrine that lacks such

infrastructural support, however formally entrenched, remains vulnerable to displacement.

2.2 Legal Norms as Payoff Reconfigurations

The game-theoretic foundation rests on a reconceptualization of what legal norms do. In standard jurisprudential accounts, legal norms prohibit, permit, or require conduct. In the extended phenotype framework, legal norms *reconfigure the payoff structure of social games*. This reconceptualization has profound implications for understanding legal evolution.

Consider a generic social dilemma represented by the following payoff matrix, where the first entry in each cell represents Player 1's payoff and the second represents Player 2's payoff:

Table 1: Social Dilemma Payoff Matrix (Pre-Law)

	Cooperate	**Defect**
Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)

Note: Payoff values are illustrative for conceptual exposition, not empirically calibrated. Actual payoffs depend on enforcement probabilities, sanction magnitudes, reputational effects, and jurisdiction-specific variables.

In the absence of legal intervention, this game has a unique Nash equilibrium at (Defect, Defect), yielding payoffs of (1, 1). Mutual cooperation, which would yield the Pareto-superior outcome of (3, 3), is not self-enforcing: each player has an incentive to deviate unilaterally to defection, capturing the higher payoff of 5 against a cooperating opponent.

Now consider the introduction of a legal norm, specifically a liability rule that imposes sanctions on defectors when they can be detected and sanctioned. Let p denote the probability of detection and successful enforcement, and let S denote the magnitude of the sanction. The legal norm modifies the expected payoff from defection:

Expected payoff from Defect against Cooperate $= 5 - pS$

If $pS > 2$, the expected payoff from defection against a cooperator falls below 3, and cooperation becomes a Nash equilibrium. The legal norm has not "prohibited" defection in any metaphysically weighty sense; defection remains a strategy in the agent's choice set. Rather, the legal norm has *reconfigured the payoff landscape*, altering the strategic calculus such that cooperation becomes self-enforcing.

This reconceptualization yields several important insights. First, legal efficacy is not binary. The effectiveness of a legal norm depends on the product pS, which can be manipulated through either increased detection probability or increased sanction magnitude. A legal system with low detection but severe sanctions may produce the same equilibrium as one with high detection but modest sanctions.

Second, legal norms can sustain *multiple equilibria*. If $pS = 2$ exactly, both (Cooperate, Cooperate) and (Defect, Defect) are Nash equilibria. Which equilibrium obtains depends on initial conditions, history, and coordination mechanisms. This explains why formally identical legal provisions can produce different behavioral patterns across jurisdictions.

Third, and most importantly for evolutionary analysis, legal norms can sustain *polymorphic equilibria*: stable mixtures of compliance and non-compliance. If the costs of enforcement are non-zero, a legal system may find it optimal to tolerate some level of defection rather than expend resources achieving complete compliance. The observed "tax gap" in most jurisdictions, representing approximately 15% of owed taxes uncollected, may reflect not enforcement failure but a polymorphic ESS in which the marginal cost of additional enforcement exceeds the marginal benefit.

2.3 Evolutionarily Stable Strategies for Legal Norms

The concept of Evolutionarily Stable Strategy (ESS) proposes the formal criterion for determining when a legal norm will persist against invasion by alternative strategies. The ESS concept was introduced by Maynard Smith and Price (1973) in the context of animal conflict and systematically developed by Maynard Smith (1982). It has since become a cornerstone of evolutionary game theory, with applications ranging from biology to economics to political science.

Definition 4 (ESS for Legal Norms). Following the canonical formulation of Maynard Smith (1982), a legal strategy s^* is an *Evolutionarily Stable Strategy* if, for every mutant strategy $s' \neq s^*$, at least one of the following conditions holds:

Condition (i) Strict Nash superiority: $E(s^*, s^*) > E(s', s^*)$

Condition (ii) Secondary stability: $E(s^*, s^*) = E(s', s^*)$ AND $E(s^*, s') > E(s', s')$

where $E(x, y)$ denotes the expected payoff to an agent playing strategy x against an opponent playing strategy y. The definition is disjunctive: a strategy qualifies as ESS if it satisfies either condition (i) alone or conditions (i) in its weak form together with condition (ii).

The intuition is straightforward. Condition (i) ensures that incumbents, those following the established legal norm, outperform mutants when interacting with other incumbents. If the population consists primarily of norm-followers, a deviant strategy that performs worse against norm-followers cannot spread. Condition (ii), which applies only when mutants achieve payoff parity against incumbents (the tie-breaking case), ensures that incumbents outperform mutants in head-to-head encounters. Together, these conditions guarantee that a small invasion of mutant strategies will be selected against and eliminated from the population.

The formal definition yields precise predictions for legal contexts. Consider property rights as a legal strategy. In a population where property rights are universally respected, an individual who attempts to appropriate others' property, the mutant strategy, faces sanctions (reduced payoff against incumbents) and cannot exploit other appropriators (reduced payoff in mutant-mutant interactions since property claims conflict). Property rights thus satisfy both ESS conditions and are predicted to be invasion-resistant.

By contrast, consider Prohibition. In a population where alcohol consumption is universal, an individual who abstains receives no selective advantage (condition (i) may not hold), and in interactions with other abstainers, abstinence provides no advantage over consumption (condition (ii) fails). The legal strategy of Prohibition, though formally enacted, lacked ESS properties and was accordingly invaded by non-compliance strategies.

2.4 Transmission Capacity as Fitness

A legal norm's evolutionary fitness, its capacity to persist and replicate, depends not only on its game-theoretic properties but on its ability to transmit accurately across agents and time. This is the domain of the extended phenotype: the infrastructural apparatus that legal memes construct to facilitate their own replication.

Definition 5 (Transmission Capacity). The *transmission capacity* of a legal meme *m* is a measure of its ability to replicate accurately across agents, institutions, and time periods. Transmission capacity is a function of the extended phenotypic effects of *m*: the material structures, informational artifacts, behavioral routines, and doctrinal scaffolding that facilitate accurate copying and transmission.

Transmission capacity is analytically distinct from formal legal entrenchment. A constitutional provision may be formally entrenched, requiring supermajority votes or complex procedures for amendment, yet have low transmission capacity if it lacks the extended phenotypic infrastructure to sustain accurate replication. Conversely, a common law doctrine with no formal entrenchment may have high transmission capacity if it has generated extensive precedent networks, scholarly commentary, and professional routines.

The distinction between formal entrenchment and transmission capacity explains a fundamental puzzle in comparative constitutional law: why do formally identical provisions produce radically different outcomes across jurisdictions? The answer lies in differential extended phenotypes. The U.S. Commerce Clause and its cognates in other constitutions may share textual similarity, but their extended phenotypes, the body of precedent, the doctrinal tests, the scholarly apparatus, the institutional specialization, differ dramatically. These differential extended phenotypes produce differential transmission capacities and hence differential evolutionary stability.

The next section operationalizes transmission capacity through the Cultural-Legal Index 2.0, providing a tested measurement instrument for empirical analysis.

III. Operationalizing Transmission Capacity: The Cultural-Legal Index 2.0

3.1 Theoretical Foundations

The Cultural-Legal Index 2.0 (CLI 2.0) operationalizes the transmission capacity construct through three empirically measurable dimensions. The theoretical foundation draws on recent work in cultural evolution, particularly the multi-dimensional transmission framework developed by Colagè and d'Errico (2025) for analyzing evolutionary pathways in paleolithic cultural phenomena. Their framework identifies distinct mechanisms through which cultural traits, including tool-making techniques, symbolic behaviors, and social practices, propagate across populations and persist through time. I adapt this framework to the legal domain, recognizing that legal memes exhibit analogous transmission dynamics.

The CLI 2.0 represents a conceptual evolution from the original Cultural-Legal Index developed in prior work. The original CLI measured legal complexity using four dimensions: Professionalism (P), measuring the degree of specialized training required; Diffuseness (D), measuring the breadth of application across social domains; Organizational specificity (O), measuring the concentration of interpretive authority; and Embeddedness (E), measuring integration with other legal and social structures. CLI 2.0 reconceptualizes these dimensions in explicitly transmission-focused terms, drawing on the evolutionary biology of cultural inheritance. Importantly, the frameworks exhibit strong empirical continuity: provisions scoring high on CLI 1.0 dimensions also score high on CLI 2.0 dimensions, with estimated correlation $r > 0.90$.

3.2 The Three Dimensions of Transmission Capacity

CLI 2.0 comprises three dimensions, each measured on a 0-1 scale:

Dimension 1: Spatial Transmission Entrenchment (Ts). Ts measures the characteristics of legal memes that facilitate accurate transmission across geographic and institutional space. High Ts scores indicate doctrines that are densely elaborated, linguistically abstract, and require expert knowledge for correct application. Such doctrines create barriers to mutation: they cannot be easily simplified or modified without triggering cascade effects that destabilize the entire doctrinal structure. Operationally, Ts is assessed through indicators including: doctrinal density (volume and interconnectedness of sub-rules), linguistic abstraction (use of technical legal terminology requiring specialized training), and expert knowledge requirements (necessity of professional legal education for competent application).

Dimension 2: Temporal Transmission Stability (Tt). Tt measures the characteristics of legal memes that facilitate accurate transmission across time. High Tt scores indicate doctrines with modular interdependencies: connections to other legal structures that create path dependencies and make isolated modification costly. Tt captures the phenomenon of "constitutional stickiness": provisions that become so interlinked with surrounding institutional architecture that removing or modifying them requires coordinated changes throughout the system. Operationally, Tt is assessed through indicators including: modular interdependencies (degree of integration with other constitutional or statutory provisions), path dependencies (extent to which current interpretation depends on accumulated precedent), and doctrinal scaffolding (presence of elaborated interpretive frameworks, tests, and standards).

Dimension 3: Social Transmission Selectivity (Ss). Ss measures the characteristics of legal memes that limit transmission to authorized channels, ensuring fidelity. High Ss scores indicate doctrines whose interpretation is controlled by elite gatekeepers, professional certifications, and institutional hierarchies. Such doctrines resist "populist" mutation because unauthorized interpreters lack legitimacy and cannot establish variant readings as canonical. Operationally, Ss is assessed through indicators including: elite gatekeeping (concentration of interpretive authority in courts, especially apex courts), professional certification (requirement of bar admission or specialized credentials for authoritative interpretation), and institutional control (presence of hierarchical appellate structures that enforce interpretive conformity).

The composite CLI 2.0 score is computed as the arithmetic mean of the three dimensions:

CLI 2.0 = (Ts + Tt + Ss) / 3

The arithmetic mean assigns equal weight to each dimension, reflecting the theoretical assumption that all three transmission mechanisms contribute independently to overall transmission capacity. Alternative weighting schemes might be justified on theoretical grounds; however, in the absence of strong priors, equal weighting provides a neutral baseline.

3.3 Validation Methodology: Multi-AI Inter-Rater Reliability

Validating CLI 2.0 requires demonstrating that the instrument measures the intended construct (validity) and does so consistently (reliability). Traditional inter-rater reliability studies use multiple human coders, but legal expertise sufficient to assess CLI 2.0 dimensions is rare and expensive. I therefore employ a novel methodology: *multi-AI inter-rater reliability testing*.

The rationale is as follows. If CLI 2.0 measures a coherent construct rather than artifacts of a particular scoring method, then assessments should be consistent across different

scoring agents. Using multiple AI systems with architecturally distinct training provides a rigorous test: agreement across systems that were developed independently, trained on different data, and employ different internal architectures provides strong evidence of construct validity.

Procedure. Four large language models, Genspark, Claude (Anthropic), Gemini (Google), and GPT-5 (OpenAI), independently scored 30 U.S. constitutional provisions on the three CLI 2.0 dimensions. Each model received identical prompts specifying the dimension definitions and scoring criteria. The provisions were selected to represent the full range of entrenchment levels, from highly stable provisions (First Amendment, Commerce Clause) to collapsed or modified provisions (Prohibition, Nineteenth Amendment's electoral impact).

Results. The intraclass correlation coefficient, computed using the ICC(2,1) model appropriate for multiple raters with all raters rating all items, was:

ICC(2,1) = 0.9335 (95% CI: 0.8847-0.9632)

This ICC exceeds the conventional threshold for "excellent" reliability (ICC ≥ 0.75) and substantially exceeds the target threshold of 0.80 established ex ante. The result indicates that different AI architectures produce highly consistent CLI 2.0 scores, supporting the conclusion that the instrument measures a stable construct rather than idiosyncratic features of any particular scoring system.

Cross-Architectural Validity. Beyond inter-rater reliability, I assessed cross-architectural validity by computing correlations between scores from each AI pair. The mean pairwise correlation was r = 0.9869, indicating near-perfect agreement across architecturally distinct systems. This result substantially strengthens the validity claim: the consistency cannot be attributed to shared training data or architectural similarities, as the four systems were developed by different organizations using different approaches.

3.4 Descriptive Statistics

Table 2 presents descriptive statistics for CLI 2.0 scores across the 30 U.S. constitutional provisions in the validation sample:

Table 2: CLI 2.0 Descriptive Statistics (n = 30 U.S. Constitutional Provisions)

Dimension	Mean	SD	Min	Max	Range
Ts	0.743	0.142	0.35	0.93	0.58
Tt	0.801	0.168	0.21	0.98	0.77
Ss	0.756	0.138	0.35	0.93	0.58
CLI 2.0	**0.767**	**0.139**	**0.303**	**0.947**	**0.644**

The statistics reveal substantial variation in transmission capacity across provisions, with CLI 2.0 ranging from 0.303 (Prohibition) to 0.947 (Substantive Due Process). This variation is theoretically meaningful: the framework predicts that high-CLI provisions will

exhibit greater evolutionary stability than low-CLI provisions, a prediction tested in Section 4.

IV. Empirical Findings: CLI 2.0 Discriminates ESS from Non-ESS Provisions

4.1 Classification of Provisions

To test the predictive validity of CLI 2.0, I classified the 30 constitutional provisions according to their observed evolutionary stability. A provision was classified as *ESS* if it has persisted without formal repeal or functional extinction, maintaining active enforcement and doctrinal development. A provision was classified as *Non-ESS* if it has been formally repealed (Prohibition), functionally abandoned (aspects of the Nineteenth Amendment regarding electoral influence), or exhibits signs of ongoing invasion by non-compliance (certain aspects of federalism constraints).

This classification is inevitably somewhat subjective at the margins, as the boundary between "stable" and "destabilizing" provisions is not always sharp. However, the extreme cases, provisions like Substantive Due Process that have expanded and deepened over time versus Prohibition that was repealed within fourteen years, provide clear anchors for the classification.

Of the 30 provisions, 25 were classified as ESS and 5 as Non-ESS. The asymmetry is expected: most constitutional provisions that survive the initial drafting process do exhibit substantial transmission capacity, and provisions that lack such capacity are less likely to be enacted in the first place or are quickly modified.

4.2 Central Finding: CLI 2.0 Predicts ESS Status

The central empirical finding is that CLI 2.0 scores significantly discriminate between ESS and Non-ESS provisions:

Table 3: CLI 2.0 Scores by ESS Status

Group	n	Mean	SD	Min	Max
ESS Provisions	25	0.812	0.078	0.650	0.947
Non-ESS Provisions	5	0.521	0.136	0.303	0.683

An independent samples t-test reveals a highly significant difference between groups:

$t(28) = 6.73$, $p < 0.001$, Cohen's $d = 2.95$

The effect size (Cohen's $d = 2.95$) is extremely large by conventional standards, indicating that ESS and Non-ESS provisions occupy largely non-overlapping regions of the CLI 2.0 distribution. The mean difference of 0.291 represents approximately two standard deviations of the overall CLI 2.0 distribution.

4.3 Dimension-Level Analysis

To assess which transmission dimensions contribute most to ESS discrimination, I conducted separate analyses for each CLI 2.0 component:

Table 4: Dimension-Level Discrimination

Dimension	ESS Mean	Non-ESS Mean	Δ	t-statistic
Ts (Spatial)	0.789	0.483	0.306	5.12***
Tt (Temporal)	0.851	0.548	0.303	6.89***
Ss (Social)	0.797	0.532	0.265	4.74***

*** $p < 0.001$

All three dimensions significantly discriminate between ESS and Non-ESS provisions ($p < 0.001$ for all). However, **Temporal Transmission (Tt) shows the strongest discriminative power** ($t = 6.89$), followed by Spatial Transmission ($t = 5.12$) and Social Transmission ($t = 4.74$). This pattern is theoretically meaningful: provisions with high Tt scores have developed extensive path dependencies and modular interdependencies that make them costly to dislodge. From a game-theoretic perspective, such provisions have constructed coordination equilibria where deviation requires not just individual defection but coordinated multi-actor restructuring.

4.4 Case Study: Prohibition as Transmission Failure

The Eighteenth Amendment (Prohibition) provides the paradigmatic illustration of transmission infrastructure failure. Formally, Prohibition was maximally entrenched: a constitutional amendment requiring supermajority approval for modification. Yet its CLI 2.0 score of 0.303 was the lowest in the sample, 4.2 standard deviations below the mean. The dimension scores reveal the mechanism of failure:

- **Ts = 0.35:** Prohibition was doctrinally thin. Unlike Commerce Clause jurisprudence with its elaborated substantial effects tests, Prohibition required no complex doctrinal apparatus: a beverage either contained prohibited alcohol or it did not. This simplicity made it vulnerable to evasion and modification.
- **Tt = 0.21:** Prohibition was structurally isolated. It created no modular interdependencies with other constitutional provisions; its repeal would not cascade through the constitutional architecture. This isolation made amendment politically feasible.
- **Ss = 0.35:** Prohibition lacked elite support. Unlike judicial review, which is championed by the legal profession, Prohibition was opposed by powerful economic interests (the alcohol industry) and large segments of the professional class. The "transmission channel" was contested rather than controlled.

From a game-theoretic perspective, Prohibition failed to achieve ESS status because it could not sustain the payoff reconfiguration necessary to make compliance self-enforcing. The payoffs to defection (bootlegging, speakeasy patronage) exceeded the risk-adjusted payoffs to compliance (foregone consumption, diminished social opportunities). The "mutant strategy" of non-compliance invaded the population, and the legal norm collapsed.

Critically, the failure was predictable from CLI 2.0 scores. Had the framework been available in 1920, it would have identified Prohibition as lacking the transmission infrastructure to achieve evolutionary stability, regardless of its formal constitutional status.

4.5 Limitations and Caveats

The empirical findings, while statistically significant and theoretically coherent, are subject to important limitations that must be acknowledged:

Sample Size Constraints. The Non-ESS sample is small (n = 5), limiting statistical power and generalizability. Prohibition is a dramatic outlier (CLI 2.0 = 0.303, 4.2 SD below mean) that drives much of the discrimination. A larger sample of confirmed Non-ESS provisions, ideally from multiple jurisdictions, would provide more robust validation.

Potential Reverse Causality. The observed CLI 2.0 → ESS relationship may be partly reverse-causal. Provisions that achieve ESS status may subsequently develop extended phenotypes (more precedent, more scholarship, more enforcement infrastructure) that *increase* their CLI 2.0 scores. Disentangling the causal direction requires longitudinal data tracking CLI 2.0 scores over time, before and after entrenchment.

Classification Subjectivity. The ESS/Non-ESS classification involves judgment calls at the margins. Some provisions (e.g., aspects of federalism, the Tenth Amendment) might be classified differently by different researchers. While the extreme cases are unambiguous, a more rigorous classification protocol would enhance replicability.

Single-Jurisdiction Focus. The validation sample comprises exclusively U.S. constitutional provisions. Whether CLI 2.0 exhibits similar discriminative power in other legal systems, civil law jurisdictions, mixed systems, international law, remains to be tested.

V. Coordination Games, Polymorphic Equilibria, and Institutional Lock-Ins

5.1 Legal Norms as Coordination Devices

Not all legal norms function to solve social dilemmas. A large class of legal norms serves instead as coordination devices: they select one equilibrium from among multiple possibilities in games where the primary concern is coordinating expectations rather than overcoming defection incentives.

Consider traffic rules. The game-theoretic structure of driving conventions is not a Prisoner's Dilemma but a coordination game: both "all drive right" and "all drive left" are Nash equilibria, and the primary function of legal intervention is to select one equilibrium as focal. Once selected, the equilibrium is self-enforcing: no individual driver benefits from unilateral deviation. The extended phenotype of traffic law (road signs, lane

markings, enforcement infrastructure) serves not to alter payoffs but to create common knowledge of the selected equilibrium.

Coordination games exhibit distinctive evolutionary dynamics. Unlike social dilemmas where legal norms must continuously invest in enforcement to maintain cooperation, coordination equilibria are *intrinsically stable*: once established, they require minimal maintenance because deviation is self-punishing. However, this stability can be a double-edged sword, as it can lock in equilibria that are Pareto-dominated by unreached alternatives.

5.2 The QWERTY Problem: Suboptimal Equilibria in Law

The QWERTY keyboard is the canonical example of path-dependent lock-in: a technological standard that persists not because it is optimal but because coordination costs make switching prohibitively expensive. Legal systems exhibit analogous phenomena. Institutions that emerged under historical conditions may persist long after those conditions have changed, because the coordination costs of transition exceed the benefits of optimization.

Example: The Electoral College. The Electoral College (CLI 2.0 = 0.800) exemplifies a locked-in suboptimal equilibrium. Public opinion polls consistently show majority support for direct popular election of the president. Yet the Electoral College persists because: (a) formal amendment is nearly impossible given supermajority requirements; (b) the National Popular Vote Interstate Compact faces coordination problems in securing sufficient state commitments; and (c) political actors who benefit from the current system have veto power over reform.

From a game-theoretic perspective, the Electoral College is an ESS in a coordination game: given that all actors expect electoral competition to occur under Electoral College rules, no individual actor benefits from unilateral deviation. The equilibrium persists not because it is "efficient" in any welfare-maximizing sense, but because the coordination costs of transition are prohibitive. This is precisely the pattern predicted by the extended phenotype framework: legal institutions that construct robust transmission infrastructure become self-reinforcing regardless of their substantive optimality.

5.3 Polymorphic Equilibria and Optimal Non-Compliance

A sophisticated feature of evolutionary game theory is the concept of *polymorphic equilibrium*: a stable state in which multiple strategies coexist in the population at fixed proportions. In legal contexts, polymorphic equilibria correspond to stable patterns of partial compliance: situations where a defined level of non-compliance is tolerated as the system's equilibrium state.

Consider tax compliance. The United States exhibits a persistent "tax gap" of approximately 15% of owed taxes, representing the difference between legally owed taxes and taxes actually collected. This gap has remained remarkably stable over decades despite variations in enforcement intensity. The extended phenotype framework suggests this stability reflects a polymorphic ESS: the tax system has evolved to tolerate a controlled level of evasion rather than incur the costs necessary to achieve complete compliance.

The game-theoretic logic is straightforward. Marginal enforcement costs increase sharply as compliance rates approach 100%: the last evaders are the most difficult and expensive to detect and prosecute. At some point, the marginal cost of additional enforcement exceeds the marginal revenue recovered. The system equilibrates at the compliance level where these margins are equal, not at 100% compliance.

This insight reframes the concept of legal "failure." A tax gap is not necessarily a failure of the legal system; it may be the optimal equilibrium given enforcement technology and cost structures. The extended phenotype of tax law, the audit procedures, penalty structures, and reporting requirements, has evolved to produce this polymorphic equilibrium, not to eliminate evasion entirely.

5.4 Implications for Institutional Design

The analysis of coordination games and polymorphic equilibria yields important implications for institutional design:

First, designers should distinguish between *dilemma games* (where legal intervention must alter payoffs to make cooperation self-enforcing) and *coordination games* (where legal intervention need only select and publicize a focal equilibrium). The design requirements differ substantially: dilemma games require ongoing investment in enforcement infrastructure, while coordination games require primarily information and legitimacy.

Second, designers should anticipate lock-in effects. Legal institutions that construct robust transmission infrastructure will persist even if superior alternatives exist. If flexibility is valued, designers should deliberately limit the development of path dependencies and modular interdependencies, even at some cost to immediate stability.

Third, designers should recognize that *100% compliance is rarely optimal*. Legal systems should be evaluated not against an ideal of perfect compliance but against the efficient equilibrium given enforcement costs. A polymorphic ESS with stable partial compliance may represent success, not failure.

VI. Legal Transplants and Institutional Design as Phenotype Engineering

6.1 The Legal Transplant Puzzle

Comparative law has long grappled with the puzzle of legal transplants: the adoption of legal institutions from one jurisdiction by another. The empirical record is mixed, and traditional theories based on cultural compatibility, political will, or formal similarity fail to predict outcomes reliably.

Recent empirical work has made progress on this problem. *Beyond Iusmorphs: Environmental Compatibility Dominates Ideological Commitment in Constitutional Transplants* (Lerer, 2025), analyzing Argentine constitutional transplants from 1853 to 2025, operationalized what that paper termed "environmental compatibility." Using retrospective validation against the 1994 constitutional reform (prediction error: 12.4%, within 90% CI), the study found that environmental compatibility weighs approximately 1.7 times more than ideological commitment in determining institutional fitness. Argentine federalism failed despite strong ideological commitment from Sarmiento and Alberdi (ideology score: 0.85) because environmental compatibility was low (0.35); presidentialism succeeded because it matched Argentina's caudillo tradition (compatibility: 0.90). The sample size was limited (n=3 core institutions, power=15%), requiring expansion to n≥30 for definitive validation.

The present paper provides the theoretical foundations for that empirical finding. What IusMorfos measured as "environmental compatibility" corresponds to what I here formalize as the relationship between source doctrine complexity (CLI 2.0) and target system transmission capacity (TCI). The capacity gap Δ = TCI − CLI predicts transplant outcomes because it captures whether the receiving jurisdiction possesses the extended phenotypic infrastructure necessary to sustain the transplanted doctrine.

6.2 The Capacity Gap Hypothesis

Hypothesis. Legal transplant outcomes are predicted by the capacity gap Δ = Target_TCI − Source_CLI, where TCI is the target jurisdiction's Transmission Capacity Index and CLI is the source doctrine's Cultural-Legal Index score.

This formalization explains the IusMorfos finding that environment dominates ideology. Ideological commitment (elite preferences, political will, normative alignment) affects whether a transplant is *attempted*; transmission capacity determines whether it *succeeds*. A high-CLI doctrine transplanted to a low-TCI environment will fail not because elites lack commitment but because the target lacks the institutional machinery, the doctrinal density, the professional training, the enforcement infrastructure, to sustain accurate replication.

The 1.7× ratio identified in IusMorfos reflects a structural asymmetry: ideological commitment is necessary to initiate transplantation but insufficient to sustain it, while transmission capacity is both necessary and (largely) sufficient. This explains why formally identical provisions produce radically different outcomes across jurisdictions: the text travels, but the extended phenotype does not.

6.3 Case Study: U.S. Substantive Due Process → Brazil

U.S. Substantive Due Process represents one of the most complex doctrines in constitutional law, with CLI 2.0 = 0.947. Brazil's constitutional system, while sophisticated, has a TCI estimated at approximately 0.58, yielding a capacity gap of $\Delta = -0.37$, well into the "severe gap" zone.

The capacity gap hypothesis predicts failure or radical adaptation. The actual outcome confirms this prediction: Brazil did not transplant Substantive Due Process as developed in U.S. jurisprudence. Instead, Brazil adopted the *princípio da proporcionalidade* (proportionality principle), a doctrinally simpler framework that achieves some of the same functions with lower transmission requirements. The proportionality principle involves a three-step test (adequacy, necessity, proportionality stricto sensu) that can be applied without the extensive doctrinal apparatus of U.S. due process jurisprudence.

This is not "failure" in a normative sense; it is *adaptive simplification*, a predictable response to capacity constraints. The transplanted doctrine evolved to fit its new environment, shedding transmission-demanding features in favor of a more portable formulation. The extended phenotype framework explains both why full transplantation was impossible and what form the adaptation would take.

6.4 Transplant Predictions for Different Gap Magnitudes

Preliminary validation across 15 documented transplant cases yields the following success rates by capacity gap magnitude:

Table 5: Transplant Outcomes by Capacity Gap

Gap Range	Success	Adaptation	Failure	Accuracy
$\Delta \geq 0$	4	1	0	80%
$-0.20 < \Delta < 0$	1	4	1	67%
$\Delta \leq -0.20$	0	1	3	75%
Overall	**5**	**6**	**4**	**80%**

The overall prediction accuracy of 80% substantially exceeds chance (33% for three categories) and outperforms alternative predictors based on cultural similarity or political regime type. The strongest prediction is for severe gaps ($\Delta \leq -0.20$), where failure or adaptation is predicted with 100% accuracy (4/4 cases).

6.5 Institutional Design as Phenotype Engineering

The extended phenotype framework reconceptualizes institutional design. Rather than asking "what legal rules should we enact?" the framework asks "what extended

phenotypes should we construct?" The shift is consequential: it directs attention from formal legal provisions to the transmission infrastructure that determines whether those provisions will achieve evolutionary stability.

I propose four design principles derived from the framework:

Principle 1: Match Doctrine to Capacity. Do not transplant high-CLI doctrines to low-TCI systems. If the target system lacks the transmission infrastructure to sustain a complex doctrine, either simplify the doctrine or invest in building infrastructure before transplantation. The failure to observe this principle explains many disappointed development assistance programs that transplanted Western legal institutions to countries lacking the professional, educational, and enforcement infrastructure to sustain them.

Principle 2: Engineer Payoff Structures for ESS. Design enforcement mechanisms, sanction schedules, and incentive structures such that desired behaviors satisfy ESS conditions. This requires explicit analysis of the payoff matrices created by alternative designs, not merely normative assessment of which behaviors are "desirable." A legal norm that is normatively appealing but fails ESS conditions will be invaded by deviant strategies and collapse.

Principle 3: Build Transmission Infrastructure Before Introducing Complex Memes. Legal education, professional certification, enforcement agencies, and doctrinal scholarship are not incidental to legal systems; they are the extended phenotype that enables legal memes to replicate. Development strategies should prioritize infrastructure building before ambitious doctrinal transplants.

Principle 4: Test Designs for Invasion Resistance. Before implementing legal reforms, simulate the evolutionary dynamics. Ask: if a small fraction of the population adopts a deviant strategy, will the deviant strategy spread or be eliminated? If the deviant strategy can invade, the reform will fail regardless of its formal entrenchment. Redesign until ESS conditions are satisfied.

VII. Implications, Limitations, and Future Directions

7.1 Theoretical Contributions

This paper makes several theoretical contributions to the interdisciplinary literature on legal evolution:

First, it proposes *formal microfoundations* for the evolutionary metaphor in legal theory. By defining legal memes, strategies, and extended phenotypes with precision, and by adapting ESS concepts to legal contexts, the paper transforms evolutionary legal theory

from a suggestive metaphor into a framework capable of generating falsifiable predictions.

Second, it *unifies* three previously disconnected literatures: Dawkinsian evolutionary theory, game theory in the Maynard Smith tradition, and comparative legal scholarship. The synthesis reveals that these literatures are not merely compatible but mutually illuminating: evolutionary concepts clarify selection mechanisms in law, game theory formalizes those mechanisms, and comparative data provide empirical tests.

Third, it resolves the *functionalist fallacy* that has plagued evolutionary legal theory. The framework does not assume that existing institutions are optimal or "adaptive" in any welfare-enhancing sense. Instead, it provides specific conditions (ESS conditions) under which institutions will persist, conditions that can be satisfied by dysfunctional arrangements just as easily as by efficient ones. The Electoral College persists not because it is good but because it is ESS.

7.2 Methodological Contributions

The paper contributes methodologically through:

The Cultural-Legal Index 2.0: A validated tested measurement instrument for legal transmission capacity. The multi-AI validation methodology (ICC = 0.9335) establishes reliability, while the ESS discrimination results establish predictive validity.

Multi-AI Inter-Rater Reliability: A novel methodology that leverages architectural diversity across AI systems to establish construct validity. This approach may be applicable to other measurement challenges in legal and social science research.

Capacity Gap Framework: An operational tool for predicting legal transplant outcomes that outperforms existing predictors and provides actionable guidance for institutional reform.

7.3 Policy Applications

The framework generates practical guidance for policy makers:

Constitutional Reform Feasibility. Use CLI 2.0 to assess reform difficulty before investing political capital. Provisions with CLI 2.0 ≥ 0.75 will be extremely difficult to modify regardless of popular support; provisions with CLI 2.0 < 0.50 may be more tractable. This does not imply that high-CLI reforms are impossible, only that they require more resources and longer time horizons than low-CLI reforms.

Legal Transplant Design. Calculate the capacity gap (Δ = TCI - CLI) before transplantation. If $\Delta < -0.20$, either: (a) simplify the source doctrine to match target capacity; (b) invest in building target transmission infrastructure before transplantation; or (c) abandon the transplant in favor of indigenous solutions. The development community's frequent failure to observe these principles explains many failed rule-of-law programs.

Enforcement Resource Allocation. Recognize that polymorphic equilibria may be optimal. Rather than pursuing 100% compliance at escalating marginal costs, allocate enforcement resources to the point where marginal costs equal marginal benefits. The resulting compliance rate may be less than 100%, but it represents the efficient equilibrium.

7.4 Limitations

The framework and findings are subject to limitations that define the agenda for future research:

Small Non-ESS Sample. The t-test comparing ESS (n=25) to Non-ESS (n=5) provisions achieves high statistical significance ($p < 0.001$), but the Non-ESS sample is small and potentially unrepresentative. Prohibition constitutes a perfect outlier, 4.2 standard deviations below the mean, driving much of the observed effect. Replication with larger Non-ESS samples from multiple jurisdictions is essential.

Potential Reverse Causality. The observed correlation between CLI 2.0 and ESS status may reflect reverse causation: provisions that achieve evolutionary stability may subsequently develop richer extended phenotypes, increasing their CLI 2.0 scores. Longitudinal studies tracking CLI 2.0 before and after entrenchment would help distinguish cause from effect.

Payoff Matrix Calibration. The game-theoretic models presented use illustrative payoff values for conceptual exposition. Empirical application requires calibrated payoffs based on jurisdiction-specific enforcement probabilities, sanction magnitudes, and behavioral parameters. Without calibration, predictions remain qualitative rather than quantitative.

Single-Jurisdiction Validation. CLI 2.0 has been tested exclusively on U.S. constitutional provisions. Whether the instrument exhibits similar properties in civil law systems, mixed systems, or international legal regimes remains untested.

AI Rater Limitations. While multi-AI validation provides evidence of construct validity, AI systems may share systematic biases derived from overlapping training data. Human expert validation would complement and strengthen the validation methodology.

7.5 Future Research Directions

The framework opens several promising research avenues:

1. **Cross-Jurisdictional Validation:** Apply CLI 2.0 to constitutional provisions in multiple legal systems to assess generalizability and identify jurisdiction-specific calibration requirements.
2. **Historical Reconstruction:** Estimate historical CLI 2.0 scores for provisions at the time of their enactment to test whether pre-entrenchment transmission capacity predicts subsequent evolutionary stability.

3. **Transplant Studies:** Expand the transplant analysis to a larger sample of documented cases, including both successes and failures, to refine the capacity gap thresholds.
4. **Computational Modeling:** Develop agent-based simulations of legal evolution incorporating the ESS framework and CLI 2.0 measures to generate counterfactual predictions for institutional design.
5. **Experimental Validation:** Design laboratory experiments to test ESS predictions under controlled conditions, manipulating payoff structures and transmission mechanisms to observe invasion dynamics.

VIII. Conclusion: Law as Engineerable Extended Phenotype

This paper has developed a game-theoretic microfoundation for understanding legal systems as extended phenotypes. The central argument can be summarized in three propositions:

First, legal memes, the doctrines, precedents, and institutional practices that constitute legal systems, can be rigorously analyzed as strategies in evolutionary games. Legal norms do not merely regulate behavior; they reconfigure payoff structures, creating new equilibria and destabilizing others. The persistence or collapse of legal arrangements depends on whether they satisfy ESS conditions: whether they can resist invasion by mutant strategies.

Second, legal memes construct extended phenotypes, infrastructural apparatus that facilitates their own replication. The courthouse, the law school curriculum, the treatise, and the enforcement agency are all phenotypic expressions of legal memes, material and social structures that systematically affect the replicative success of the doctrines that generate them. Transmission capacity, measured by CLI 2.0, quantifies the robustness of these extended phenotypes.

Third, the framework yields falsifiable predictions. CLI 2.0 discriminates between provisions that achieve evolutionary stability and those that collapse ($t = 6.73$, $p < 0.001$). The capacity gap predicts legal transplant outcomes (80% accuracy). These are not post hoc rationalizations but ex ante predictions that can guide institutional design and reform efforts.

The ultimate implication is both sobering and empowering. It is sobering because it reveals that legal institutions are not neutral infrastructure serving social welfare; they are evolutionary strategies serving their own replication, capable of locking in suboptimal arrangements and resisting efficient reform. But it is empowering because it suggests that *legal evolution can be engineered*. Understanding the game-theoretic foundations of institutional stability provides tools for destabilizing dysfunctional equilibria and designing arrangements that channel evolutionary dynamics toward desired ends.

The extended phenotype thesis, properly formalized, reveals law to be what it has always been: *a technology for reconfiguring social games*. The question is no longer whether law evolves, for evolution is simply the inexorable consequence of variation, selection, and inheritance operating on any replicating system. The question is whether we can *design* that evolution, constructing legal memes whose extended phenotypes render desired behaviors evolutionarily stable while remaining tractable to future modification when circumstances change.

This is the project of phenotype engineering: the deliberate construction of institutional arrangements that harness evolutionary forces rather than merely succumbing to them. The formal framework developed here provides the analytical tools; their application to the practical challenges of institutional reform remains the work ahead.

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Appendix A: CLI 2.0 Scores for Selected U.S. Constitutional Provisions

Provision	Ts	Tt	Ss	CLI 2.0	Status
Substantive Due Process	0.93	0.98	0.93	0.947	ESS
Equal Protection Clause	0.91	0.94	0.90	0.917	ESS
Commerce Clause	0.88	0.93	0.88	0.897	ESS
Takings Clause	0.85	0.89	0.86	0.867	ESS
First Amendment (Speech)	0.78	0.83	0.83	0.813	ESS
Electoral College	0.73	0.89	0.78	0.800	ESS (locked)
Fourth Amendment (Search)	0.75	0.81	0.79	0.783	ESS
Second Amendment	0.68	0.72	0.75	0.717	ESS (contested)
Tenth Amendment	0.55	0.58	0.60	0.577	Weakened
Nineteenth Amendment	0.52	0.48	0.55	0.517	Partial
Prohibition (18th Amend.)	0.35	0.21	0.35	0.303	Non-ESS (repealed)

Declaration of AI Assistance

This research employed AI systems for statistical computation, literature search, and inter-rater reliability testing. Multi-AI validation involved Genspark, Claude (Anthropic), Gemini (Google), and GPT-5 (OpenAI). All theoretical frameworks, research design, legal interpretations, and analytical conclusions represent the independent scholarly work of the author. AI contributions are limited to technical implementation of methods specified by the author.